

To note: The entire system is extremely costly (e.g., 32 electrodes around 1500€ / the entire system between 15-30k€ / caps 520€).

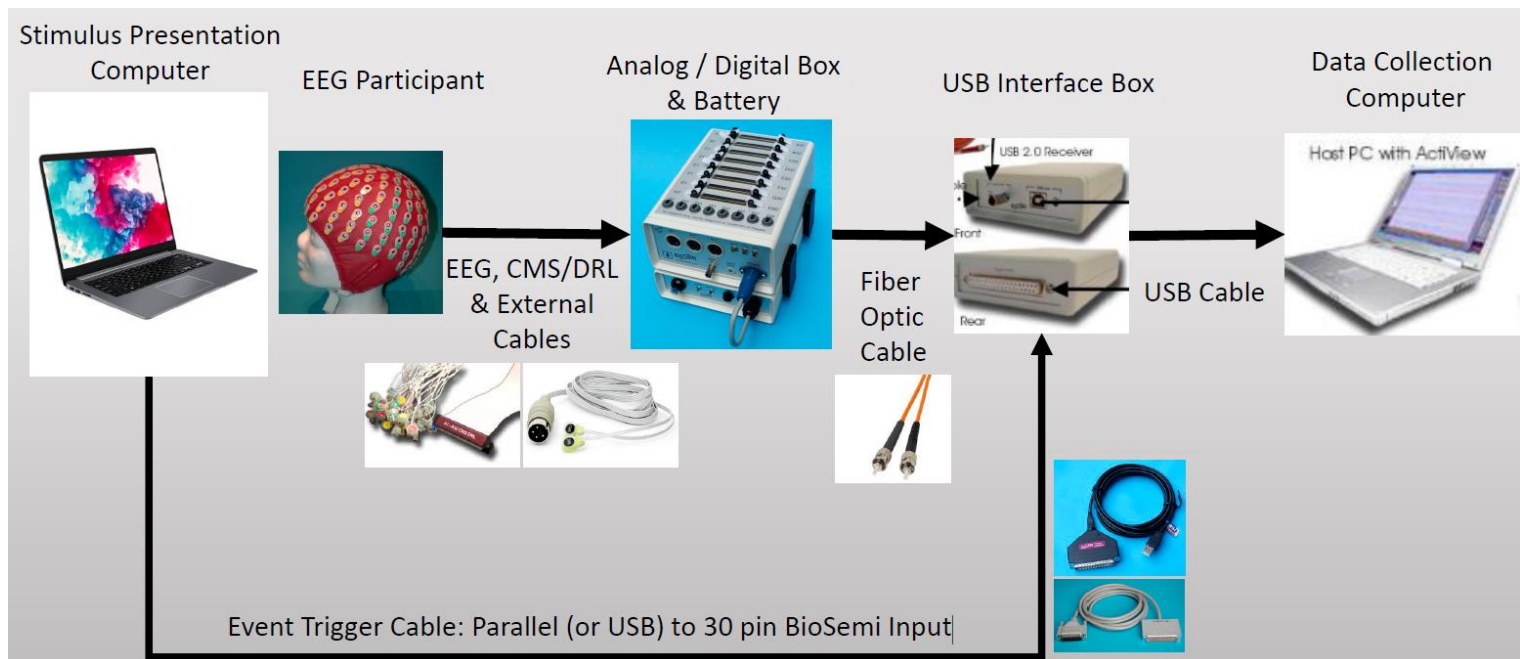
BEFORE THE PARTICIPANT ARRIVAL :

Open actiview on the data collection computer:

Load the config file (if not already done)

CAREFULL ! Fiber cable is very fragile do not fold it (or walk on it) - avoid putting any pressure on it

Connect all the cables :



Prepare EEG :

Connect the electrodes (set of electrodes + depends on the study: externals)

Fill 2 syringes of gel

Depends on the study : prepare some stickers for externals

Depends on the EEG : connect the CMS/DRL external (some eeg have it directly on the set of electrodes)

Check the battery of the EEG, at least 3 bars

Let your supervisors know if you have any problems with your equipment.

The same applies if you run out of consumables (stickers, gel, etc.): plan ahead for the next supply.

INSTALLATION EEG ON THE PARTICIPANT :

Optional : connect the external (e.g., mastoids)

Measure the head to choose a cap (better to choose smaller cap than larger in case of doubt) - Put the cap and centered it on Cz

Attach chin strap (be sure participants are comfortable, do not strangle them!)

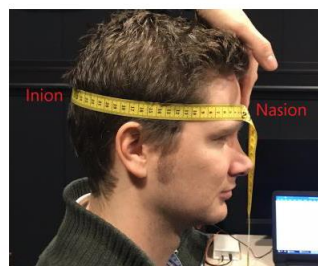


Figure 5. Measure head circumference

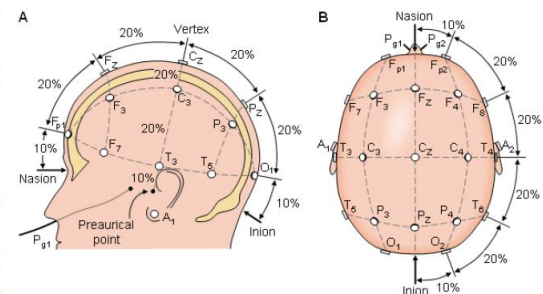


Figure 6. Make sure the cap is properly centered

Put gel inside the gaps :

gently touch the skin with the syringe, then make some rotative movement to pull the hair, finally press the syringe to put gel while bringing it up to fill the gap. (Better to put in not enough gel, we can easily add, but we cannot remove it if too much).

Plug the electrodes in the gaps of the cap.

Better to lose 10 minutes to have a good signal during the experiment, instead of throwing away the data (= the passation for nothing ~ 1h lost) and/or losing time (a lot of time) to clean the messy data.

The screenshot displays the BIOPAC MP150 software interface, which is used for recording and analyzing neural data. The main window shows a multi-channel EEG recording with waveforms for various channels (F7, F8, F9, etc.). The left sidebar contains settings for 'Y-axis Voltage' (100 uV), 'Lowpass' (100 Hz), 'Highpass' (0.5 Hz), 'Channels' (All 32), and 'Reference' (Avg 14). The right sidebar shows 'BCF File' settings, 'Sample Rate' (512 Hz), 'Sampling Rate' (1000 Hz), 'Gain Factor' (A1-B2 0.4), and 'Filter' settings (Low 1 Hz, High 30 Hz, Band 1 Hz, Band 30 Hz). The bottom status bar shows 'Tag 1-4' and 'Tag 9-16'.

IMPORTANT: Be careful to avoid electrolyte bridges and particularly between CMS and/or DRL and active electrodes. An electrolyte bridge (short) between any active electrode and DRL will result in very high noise in the signal measured from the active electrode. An electrolyte bridge between any active electrode and CMS will result in a flat-line (no voltage can be measured between two shorted contacts).

- Start by checking the CMS / DRL (if there's not enough gel, put some back in, but be very careful with the bridge).

Bad CMS / DRL Connection



Almost always the solution is to add more gel to CMS / DRL

** Could also indicate a bad electrode. **



Will be blinking

During the experiment the gel could dry resulting in bad signals for some electrodes, keep an eye on the signal to spot when it happens in order to arrange it (put more gel for this electrode most of the time)

Instruct and check that the participants do not move too much the head, it produces noise in the signal, also check that they are decontracted, any muscular contraction produces noise in the signal.

- use a towel to gently clean the electrodes (set of electrodes and externs)
- Remove the gel from the gaps in the cap (e.g., with a brush or a syringe)

Keep the connectors far away from water and metal !!!

- inside the syringes and then press the syringe to expulse the water. Do it at least twice. You can use the water jet of the syringes on the electrodes and the caps to start cleaning them
- on the electrodes (KEEP THE CONNECTORS AWAY) and clean them gently
- on the cap and clean the gel inside the gaps (e.g., with a brush or a syringe).
- optional : In a container to let the electrodes and the cap 5-10 minutes inside before cleaning them properly, it helps.

